

SAKSHI SANJAY BHARAMBE

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EDUCATION

University of Southern California

Los Angeles, CA

Master of Science in Computer Science

August 2024 – May 2026

- Coursework: Analysis of Algorithms, Web Technologies, Multimodal Learning, Applied NLP, Information Retrieval and Web Search Engines, Research Methods and Analysis for User Studies, Deep Learning, Database Systems

Terna Engineering College, University of Mumbai

Mumbai, India

Bachelor of Engineering in Computer Engineering

August 2019 – June 2023

- Coursework: Artificial Intelligence, Machine Learning, Applied Data Science, Big Data Analytics, Software Engineering, Data Structures, Computer Network

SKILLS

Languages: Python, Java, TypeScript, JavaScript, SQL, C, HTML, CSS, PHP, Swift

Frameworks: React, Node.js, Express.js, Selenium, Flask, FastAPI, Bootstrap

Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe, Pandas, NumPy, NLTK, Matplotlib, XGBoost

Databases: MySQL, MongoDB, Redis

Tools and Platforms: Git, GitHub, Docker, AWS, Azure AD, VS Code, Jupyter Notebook, PyCharm, Google Colab

Engineering Concepts: Web Scraping, Automation, Multithreading, REST APIs, OOP, Data Pipelines

EXPERIENCE

USC Transportation

Los Angeles, CA

Software Developer

September 2024 – May 2026

- Automated data-intensive workflows for internal operations by developing Python scripts with Selenium and Pandas, reducing manual data entry tasks and boosting processing efficiency by 75%
- Designed a scalable web scraping and auto-fill pipeline with multithreaded execution, implementing error handling and logging to enhance data reliability and increase reservation throughput by 5 times
- Implemented Microsoft Outlook notifications into enterprise systems via Azure AD and Python-based APIs, facilitate real-time alerts and reduce communication delays by 40% across operational teams

Leap Analytics

California, USA

Software Engineer Intern

May 2025 – August 2025

- Improved Leap's existing website by adding new pages, refining UI components, and restructuring responsiveness with HTML, CSS, and JavaScript, promoting client engagement by 20%
- Constructed automated web scraping pipelines using Python and Selenium to extract structured datasets from multiple sources, decreasing data collection time by 80% and accelerating market research
- Performed data cleaning, exploratory analysis, and visualization with Pandas, NumPy, and Matplotlib, improving reporting accuracy by 25% and enabling more reliable strategy decisions

Blend Vidya

Mumbai, India

Artificial Intelligence Intern

January 2023 – June 2023

- Analyzed over 10,000+ student activity records to derive engagement patterns by applying data cleaning, displaying, and visualization leveraging Pandas, NumPy, and Matplotlib, elevated reporting accuracy by about 30%
- Integrated AI-driven insights by developing intelligent data visualization tools and feedback analytics, enhancing mentor-student interactions and increasing learner satisfaction by approximately 20%
- Led a team of 4 interns to create a predictive analytics module for course recommendation by leveraging clustering and classification models, boosting student engagement by 25% based on personalized suggestions

ACADEMIC PROJECTS

Gesture Controlled Virtual Mouse with Voice Assistant | *Computer vision, Speech Recognition*

- Engineered a real-time virtual mouse using OpenCV and MediaPipe with gesture and voice controls, enabling cursor navigation, clicks, drag-and-drop, and volume adjustment with 98% gesture recognition accuracy
- Implemented hand tracking using 21 MediaPipe landmarks and integrated speech recognition to enable hands-free system interaction, achieving 100% accuracy for core gestures across 700 tests

Multimodal Emotion Recognition System | *BERT, HuBERT, Streamlit*

- Designed a multimodal emotion recognition pipeline using BERT, HuBERT, and EfficientNet-B3 to process text, audio, and video inputs, achieving 60% accuracy through weighted late fusion
- Developed a Streamlit-based inference application that automated preprocessing, feature extraction, and emotion prediction, enabling end-to-end emotion recognition from uploaded videos through a unified inference pipeline

QueryBot | *BERT, T5, SQL*

- Built a scalable preprocessing pipeline for the Spider dataset by standardizing data cleaning, schema linking, and JSON/CSV serialization, enabling efficient training of transformer-based Text-to-SQL models
- Optimized schema-aware preprocessing workflows and interactive data visualizations, reducing query parsing errors by 30% and improving SQL generation across diverse database schemas

Gene Semantic Bridge | *BioBERT, CVAE, Generative Modeling*

- Built a KA-CVAE framework integrating BioBERT-derived gene embeddings with learnable representations, improving transcriptomic prediction for unseen CRISPR perturbations
- Validated the framework on the Norman scRNA-seq benchmark, achieving an R^2 of 0.996 while outperforming baseline models on out-of-distribution perturbation tasks